

C0859

Database Theory

Lecture 10

Normalization – Part 1

Introduction

- We want to develop a **conceptual** model that represents the **entities** of the enterprise and the **relationships** between them
- This model is **independent** of any DBMS package, hardware or the **physical** model
- Later on, it will be mapped to an **internal** model that complies with the chosen DBMS

Relational Concepts For The Design Process

- We use the **normalization** process to group data elements into **tables** that represent **entities**

Normalization

- Normalization is the process of organizing data to minimize redundancy
- Normalized data doesn't have storage anomalies
- The organizing of data is done by applying a series of rules to a data model

Normal Forms

- **0 Normal Form (0NF):** un-normalized model before the normalization rules have been applied
- **First Normal Form (1NF)**
- **Second Normal Form (2NF)**
- **Third Normal Form (3NF)**
- **Boyce Codd Normal Form (BCNF)**
- **Fourth Normal Form (4NF)**
- **Fifth Normal Form (5NF)**
- **Domain Key Normal Form (DKNF)**
- Each previous level must be applied before proceeding to the next

Normalization

- While normalization makes databases more efficient to maintain, they can also make them more complex because data is separated into many different tables
- This will require most queries to use table joins

First Normal Form (1NF)

- Reduce entities to first normal form (1NF) by removing repeating or multi-valued attributes to another, child entity
- To simplify, you cannot have multiple fields to capture multiple values
- Most common 'sins' against the first normal form are the lack of a primary key and the use of repeating columns
- Also, remove calculated fields

ONF (un-normalized)

- No rules have been applied to this model
- Note the repeating product columns
- This does not allow for more products to be added and creates cells with null values
- Also note the calculated fields: $\text{ExtPrice1} = \text{Qty1} * \text{Price1}$,
 $\text{Amount} = \text{ExtPrice1} + \text{ExtPrice2} + \text{ExtPrice3}$

Order	OrderDate	CustId	LName	FName	Prov	TaxRate	Prod1	ProdDesc1	Qty1	Price1	ExtPrice1	Prod2	ProdDesc2	Qty2	Price2	ExtPrice2	Prod3	ProdDesc3	Qty3	Price3	ExtPrice3	Amount
239	2024-07-05	1135	Black	John	ON	0.13	555	Cheese Tray	10	10.00	100.00	444	Wine Gift	15	20.00	300.00						400.00
240	2024-07-07	1135	Black	John	ON	0.13	343	Bottle Opener	15	2.00	30.00	555	Cheese Tray	20	10.00	200.00	111	Wine Glass	25	4.00	100.00	330.00
241	2024-07-08	1140	Brown	Jack	ON	0.13	231	Hat	5	7.50	37.50	123	Bat	2	25.00	50.00						87.50
243	2024-07-12	1140	Brown	Jack	ON	0.13	321	Cards	25	3.00	75.00											75.00
245	2024-07-13	1152	Mills	Mike	BC	0.12	451	Shoes	4	30.00	120.00	321	Cards	15	3.00	45.00						165.00
246	2024-07-17	1156	Levey	Kyle	MB	0.12	555	Cheese Tray	5	10.00	50.00	231	Hat	10	7.50	75.00	321	Cards	20	3.00	60.00	185.00
255	2024-07-19	1174	Little	Tony	SK	0.11	444	Wine Gift	8	20.00	160.00	12	Soccer Ball	12	10.00	120.00						280.00
256	2024-07-22	1174	Little	Tony	SK	0.11	945	Chair	2	50.00	100.00	932	Table	1	100.00	100.00	910	Couch	1	300.00	300.00	500.00
260	2024-07-23	1180	Nye	Bill	PE	0.15	800	Glasses	100	2.50	250.00											250.00

1NF

- Two tables have been created: order and product
- They are linked by the Order field
- This eliminates the repeating fields in a record
- Calculated values ExtPrice1, ExtPrice2, ExtPrice3 and Amount have been discarded

1NF

Order	OrderDate	CustId	LName	FName	Prov	TaxRate
239	2024-07-05	1135	Black	John	ON	0.13
240	2024-07-07	1135	Black	John	ON	0.13
241	2024-07-08	1140	Brown	Jack	ON	0.13
243	2024-07-12	1140	Brown	Jack	ON	0.13
245	2024-07-13	1152	Mills	Mike	BC	0.12
246	2024-07-17	1156	Levey	Kyle	MB	0.12
255	2024-07-19	1174	Little	Tony	SK	0.11
256	2024-07-22	1174	Little	Tony	SK	0.11
260	2024-07-23	1180	Nye	Bill	PE	0.15

Order	Prod	ProdDesc	Qty	Price
239	555	Cheese Tray	10	10.00
239	444	Wine Gift	15	20.00
240	343	Bottle Opener	15	2.00
240	555	Cheese Tray	20	10.00
240	111	Wine Glass	25	4.00
241	231	Hat	5	7.50
241	123	Bat	2	25.00
243	321	Cards	25	3.00
245	451	Shoes	4	30.00
245	321	Cards	15	3.00
246	555	Cheese Tray	5	10.00
246	231	Hat	10	7.50
246	321	Cards	20	3.00
255	444	Wine Gift	8	20.00
255	12	Soccer Ball	12	10.00
256	945	Chair	2	50.00
256	932	Table	1	100.00
256	910	Couch	1	300.00
260	800	Glasses	100	2.50

PK highlighted in yellow

Second Normal Form (2NF)

- Reduce first normal form entities to second normal form (2NF) by removing attributes that are not dependent on the whole primary key
- The primary key for each record must be able to determine the value for all of the other fields in the records

1NF

- Here knowledge of the Product # alone provides the product description and price without the Order#
- We must create another table for this to conform to 2NF

Order	Prod	ProdDesc	Qty	Price
239	555	Cheese Tray	10	10.00
239	444	Wine Gift	15	20.00
240	343	Bottle Opener	15	2.00
240	555	Cheese Tray	20	10.00
240	111	Wine Glass	25	4.00
241	231	Hat	5	7.50
241	123	Bat	2	25.00
243	321	Cards	25	3.00
245	451	Shoes	4	30.00
245	321	Cards	15	3.00
246	555	Cheese Tray	5	10.00
246	231	Hat	10	7.50
246	321	Cards	20	3.00
255	444	Wine Gift	8	20.00
255	12	Soccer Ball	12	10.00
256	945	Chair	2	50.00
256	932	Table	1	100.00
256	910	Couch	1	300.00
260	800	Glasses	100	2.50

2NF

- Another table has been created: order_line
- It is linked to the product table with the common field product
- Therefore each product description is entered once in the database rather than each time someone purchases the item
- The combination of Order# and Product# are needed to identify quantity

2NF

Order	Prod	Qty
239	555	10
239	444	15
240	343	15
240	555	20
240	111	25
241	231	5
241	123	2
243	321	25
245	451	4
245	321	15
246	555	5
246	231	10
246	321	20
255	444	8
255	12	12
256	945	2
256	932	1
256	910	1
260	800	100

Prod	ProdDesc	Price
12	Soccer Ball	10.00
111	Wine Glass	4.00
123	Bat	25.00
231	Hat	7.50
321	Cards	3.00
343	Bottle Opener	2.00
444	Wine Gift	20.00
451	Shoes	30.00
555	Cheese Tray	10.00
800	Glasses	2.50
910	Couch	300.00
932	Table	100.00
945	Chair	50.00

Getting Smaller

- The order table that started with 23 fields has now shrunk considerably
- 3NF will reduce this further by removing the redundant names, provinces and tax rates

Order	OrderDate	CustId	LName	FName	Prov	TaxRate
239	2024-07-05	1135	Black	John	ON	0.13
240	2024-07-07	1135	Black	John	ON	0.13
241	2024-07-08	1140	Brown	Jack	ON	0.13
243	2024-07-12	1140	Brown	Jack	ON	0.13
245	2024-07-13	1152	Mills	Mike	BC	0.12
246	2024-07-17	1156	Levey	Kyle	MB	0.12
255	2024-07-19	1174	Little	Tony	SK	0.11
256	2024-07-22	1174	Little	Tony	SK	0.11
260	2024-07-23	1180	Nye	Bill	PE	0.15

Third Normal Form (3NF)

- Third normal form is violated when a non-key field is a fact about another non-key field
- Reduce second normal form entities to third normal form (3NF) by removing attributes that are dependent on other attributes

3NF

- By creating another two tables our order table is down to 3 fields
- We've eliminated the name fields by linking the CustID to a separate customer table, and likewise for the tax rate using the province abbreviation

Order	OrderDate	CustId
239	2024-07-05	1135
240	2024-07-07	1135
241	2024-07-08	1140
243	2024-07-12	1140
245	2024-07-13	1152
246	2024-07-17	1156
255	2024-07-19	1174
256	2024-07-22	1174
260	2024-07-23	1180

3NF

- This removes the redundancy of the customer name and province each time an order is placed

CustId	LName	FName	Prov
1135	Black	John	ON
1140	Brown	Jack	ON
1152	Mills	Mike	BC
1156	Levey	Kyle	MB
1174	Little	Tony	SK
1180	Nye	Bill	PE

3NF

- Here the redundancy of the tax rate is removed by creating a table with province tax information

Prov	TaxRate
BC	0.12
MB	0.12
ON	0.13
PE	0.15
SK	0.11

Order Normalization

- The original un-normalized Order table became 5 normalized tables
 - customers
 - orders
 - order_lines
 - products
 - provinces

Third Normal Form Tables

- customers
 - customer_id -> fname, lname, province_id
- orders
 - order_id -> order_date, customer_id
- order_lines
 - order_id + product_id -> qty
- products
 - product_id -> product_description, price
- provinces
 - province_id -> tax_rate

Creating the ERD

- Draw the entities with their primary keys

orders	
PK	order_id
	order_date
	customer_id

customers	
PK	customer_id
	lname
	fname
	province_id

order_lines	
PK	order_id
PK	product_id
	qty

provinces	
PK	province_id
	tax_rate

products	
PK	product_id
	product_description
	price

Creating the ERD

- Examine each primary key to see if it appears in another table, if so it is a foreign key in the other table

orders	
PK	order_id order_date
FK	customer_id

customers	
PK	customer_id lname fname
FK	province_id

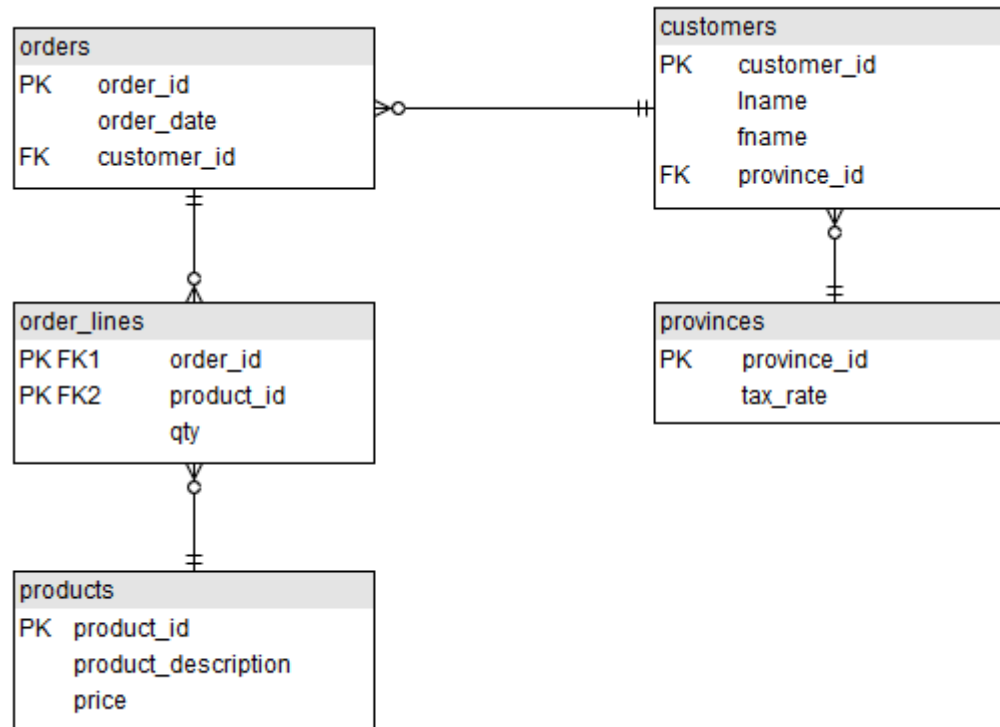
order_lines	
PK FK1	order_id
PK FK2	product_id qty

provinces	
PK	province_id tax_rate

products	
PK	product_id product_description price

Creating the ERD

- Each foreign key means there is a relationship between the two tables
- The primary key is on the “one” side and the foreign key is on the “many” side



Normalization Good News/Bad News

- The bad news is that this increases processing complexity
 - In order to get any useful information out of the database, most queries will need to join 2 or more tables
- The good news is that the data is organized and unique
 - Each fact is only stored in one place

Normalization Quick Summary

- 1NF - Eliminate Repeating Groups And Derived Attributes
 - Make a separate table for each set of related attributes, and give each table a primary key
- 2NF - Eliminate Redundant Data
 - If an attribute depends on only part of a composite key, remove it to a separate table
- 3NF - Eliminate Columns Not Dependent On Key
 - If attributes do not contribute to a description of the key, remove them to a separate table

More Examples On The Internet

- https://en.wikipedia.org/wiki/Database_normalization
- <https://docs.microsoft.com/en-us/office/troubleshoot/access/database-normalization-description>
- <http://www.bkent.net/Doc/simple5.htm>
- <http://www.troubleshooters.com/littstip/ltnorm.html>
- <http://databases.about.com/library/weekly/aa080501a.htm>
- <http://phlonx.com/resources/nf3/>

Normalization Case Study

- Pipes N Things Inc.
 - A distributor of plumbing parts and fixtures
 - Purchase from manufacturers
 - Supply retailers
 - Designing corporate data base

Purchasing Department

- Determine vital entities
- Purchase Order document is critical

List The Entities Found

Purchase Order				
Date:		2024-07-15	Vendor: Crane Plumbing	
P.O. #:		3067	572 Johnson St.	
Vendor ID:		507	Willowdale, ON	
			M2M 3E4	
Qty	Item ID	Description	Unit Price	Line Total
80	P-796	Delta Cartridges	3.50	280.00
100	P-797	Moen Cartridges	3.25	325.00
25	P-993	Kitchen Faucets	18.75	468.75
Total				1,073.75

- Hint, look for keys
- Keys will usually identify entities

A Second PO

Purchase Order

Date: 2024-07-20

P.O. #: 3068

Vendor ID: 385

Vendor: MCE Supply
12 Burris St.
Hamilton, ON
L8M 5R4

Qty	Item ID	Description	Unit Price	Line Total
20	P-796	Delta Cartridges	3.50	70.00
50	P-853	2" O Rings	1.25	62.50
Total				132.50

Convert The Purchase Order Documents Into A Two Dimensional Table

PO_No	Date	Vend_ID	Name	Address	City	Prov	Postal	Qty	Item_ID	Description	Price	Line_Total	Total
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	80	P-796	Delta Cartridges	3.50	280.00	1073.75
								100	P-797	Moen Cartridges	3.25	325.00	
								25	P-993	Kitchen Faucets	18.75	468.75	
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	20	P-796	Delta Cartridges	3.50	70.00	132.50
								50	P-853	2" O Rings	1.25	62.50	

- Results in an un-normalized table
- Multiple occurrences of data exist
- We can find cells that have more than one value
 - Item_ID, Desc, Qty, Description, Price, Line_Total

Normalization – The First Step

- Transforming an un-normalized table into one that is in First Normal Form
- **1NF** has no repeated occurrences of data items in any of its cells
- Go back to the table and put it into 1NF by repeating some of the data values and removing calculated values

Normalization – The First Step

- Repeating Values For Attributes
 - PO_No, Date, Vend_ID, Name, Address, City, Prov, Postal
- Calculated values (Line_Total, Total) removed

PO_No	Date	Vend_ID	Name	Address	City	Prov	Postal	Qty	Item_ID	Description	Price
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	80	P-796	Delta Cartridges	3.50
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	100	P-797	Moen Cartridges	3.25
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	25	P-993	Kitchen Faucets	18.75
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	20	P-796	Delta Cartridges	3.50
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	50	P-853	2" O Rings	1.25

What is the table's key?

...PO_No + Item_ID

Normalization – The Second Step

- Isolating
 - What data items are dependent on parts of the key
 - What data items are left dependent on the whole key

- Dependent on PO_No
 - Date
 - Vend_ID
 - Name
 - Address, City, Prov, Postal
- The resulting table
 - po_no -> date, vend_id, name, address, city, prov, postal

- Dependent on Item_ID
 - Description
 - Price
- The resulting table
 - item_id -> description, price

- The attribute(s) remaining as dependent on the original key is/are
 - Qty
- The third table
 - po_no + item_id -> qty

Normalization – The Third Step

- Separating any entity that deserves its own relation in the database
- Name each entity
- Look for a key field that has a dependent attribute; a key for another entity
- Is this occurring in our 2NF tables?
- If so, where?

Third Normal Form Tables

- po_headers
 - po_no -> date, vend_id
- vendors
 - vend_id -> name, address, city, postal, prov
- items
 - item_id -> description, price
- po_lines
 - po_no + item_id -> qty

Creating the ERD

- Draw the entities with their primary keys

po_headers	
PK	po_no
	date
	vend_id

vendors	
PK	vend_id
	name
	address
	city
	prov
	postal

po_lines	
PK	po_no
PK	item_id
	qty

items	
PK	item_id
	description
	price

Creating the ERD

- Examine each primary key to see if it appears in another table, if so it is a foreign key in the other table

po_headers	
PK	po_no
	date
FK	vend_id

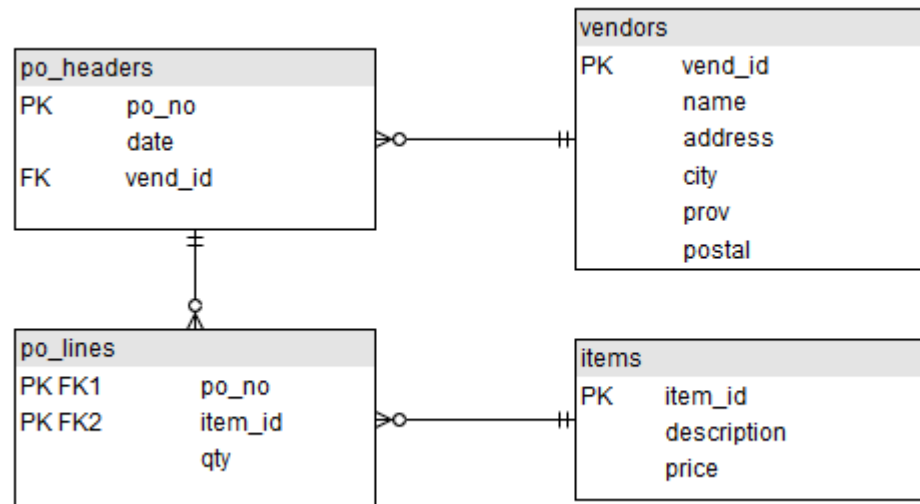
vendors	
PK	vend_id
	name
	address
	city
	prov
	postal

po_lines	
PK FK1	po_no
PK FK2	item_id
	qty

items	
PK	item_id
	description
	price

Creating the ERD

- Each foreign key means there is a relationship between the two tables
- The primary key is on the “one” side and the foreign key is on the “many” side



Storage Anomalies of the First Normal Form

- Insertion Anomaly
- Update Anomaly
- Deletion Anomaly

1NF Insertion Anomaly

PO_No	Date	Vend_ID	Name	Address	City	Prov	Postal	Qty	Item_ID	Description	Price
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	80	P-796	Delta Cartridges	3.50
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	100	P-797	Moen Cartridges	3.25
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	25	P-993	Kitchen Faucets	18.75
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	20	P-796	Delta Cartridges	3.50
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	50	P-853	2" O Rings	1.25

- Recall the first normal form again for the Purchase Order
- If the database had only this table in it, we could not set up a Vendor or Item unless we had all the PO information

1NF Update Anomaly

PO_No	Date	Vend_ID	Name	Address	City	Prov	Postal	Qty	Item_ID	Description	Price
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	80	P-796	Delta Cartridges	3.50
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	100	P-797	Moen Cartridges	3.25
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	25	P-993	Kitchen Faucets	18.75
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	20	P-796	Delta Cartridges	3.50
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	50	P-853	2" O Rings	1.25

- If the Description for an item changed, we would have to find the Description and change all occurrences

1NF Deletion Anomaly

PO_No	Date	Vend_ID	Name	Address	City	Prov	Postal	Qty	Item_ID	Description	Price
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	80	P-796	Delta Cartridges	3.50
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	100	P-797	Moen Cartridges	3.25
3067	2024-07-15	507	Crane Plumbing	572 Johnson St.	Willowdale	ON	M2M 3E4	25	P-993	Kitchen Faucets	18.75
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	20	P-796	Delta Cartridges	3.50
3068	2024-07-20	385	MCE Supply	12 Burris St.	Hamilton	ON	L8M 5R4	50	P-853	2" O Rings	1.25

- Deleting the rows for a Purchase Order causes the loss of Vendor and Item information

Storage Anomalies of the Second Normal Form

- Advantages of 2NF
- INSERTION
 - An item can be added by itself without PO information
- UPDATE
 - If the Description for an item changed, only one occurrence would be affected
- DELETION
 - deleting a row for a PO has no effect on an Item record

Storage Anomalies Of The Second Normal Form

- Disadvantages of 2NF
- INSERTION
 - Vendor cannot be set up by itself
- UPDATE
 - multiple records may need to be updated if Vendor address information changed
- DELETION
 - deleting rows for a PO causes loss of Vendor information
 - po_no -> date, vend_id, name, address, city, prov, postal

The Absence of Storage Anomalies In The Third Normal Form

- In one of the 2NF relations, we have a transitive dependency between one non-key attribute, vend_id, and other non-key attributes, name, address, city, prov and postal
- The transitivity is removed when VENDOR has its own relation in the database
 - po_no -> date, vend_id, name, address, city, prov, postal

Advantages Of Third Normal Form

- Vendor can be set up by itself
- Only one record needs updating if vendor address changes
- Deleting rows for a PO does not cause loss of vendor information
 - vend_id -> name, address, city, prov, postal

Summary

- Normalization is the process of organizing data to minimize redundancy
- Normalized data doesn't have storage anomalies
- 1NF - Eliminate Repeating Groups And Derived Attributes
- 2NF - Eliminate Redundant Data
- 3NF - Eliminate Columns Not Dependent On Key